

Jordan Vogel

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EDUCATION

Cornell University, College of Engineering, Ithaca, NY
Bachelor of Science, Mechanical Engineering GPA: 4.048
Master of Engineering, Mechanical Engineering

Expected Dec 2026
Expected May 2027

Relevant Courses: *Biofluids Mechanics, Computer-Aided Manufacture, Heat Transfer, Fluid Mechanics, System Dynamics, Mechanics of Materials, Foundations of Robotics, Advanced Product Design, Propulsion of Aircraft and Rockets, Fundamentals of Electric-Drive Vehicles, Dynamics, Statics, Mechatronics*

PROFESSIONAL EXPERIENCE

General Motors, Vehicle Optimization and Machine Learning Intern **May 2026 - Aug 2026**

- Created a live dashboard that tracks the integration status of all vehicle programs in development across the company using SharePoint List, Power BI, Power Automate, and Power Apps
- Decreased effort required to document progress and increased visibility of total vehicle status
- Trained new ML models to predict emissions of L76 engine, tripling the number of outputs the tool can provide

Cornell Electric Vehicles Project Team, Cornell University, *Mechanical Lead/Senior Advisor* **Sep 2023 - Present**

- Advise a 75+ member organization developing hyper-efficient autonomous electric vehicles for the annual Shell Eco-marathon by attending design reviews, offering expertise, and promoting best practices in automotive design
- Previously led a 25+ member mechanical subteam within the larger team
- Served as driver, placing 6th in 2026 with an efficiency of 37 mi/kWh, 84% more efficient than the previous year
- Served as team's only TRAK lathe machinist, owning the manufacturing of our spindles and driveshafts
- Owned steering system layout for 2026 vehicle
- Designed a threaded aluminum mounting plate that defines the interface for the rack and pinion, pedal box, and autonomy hardware, engineered for maximum rigidity and minimal weight
- Designed, built, and tested a gearbox enabling autonomous steering via stepper motor and seamless transition to manual control through an electromagnetic clutch; selected gear ratio and motor to achieve 5 N*m torque required
- Designed a parking brake that can keep the vehicle stable on a 20% incline
- Increased reliability of the parking brake by introducing a hydraulic line lock into the braking system and reducing the number of overall parts in the assembly

Lutron Electronics, *Manufacturing Engineering Intern* **May 2025 - Aug 2025**

- Researched lasers as a method of marking product information, specifically focusing on reducing costs, operator error, and assembly time
- Acquired and learned how to use two desktop lasers for testing, both completely new to the company
- Developed a framework to find optimal settings for a given material quickly and efficiently, varying settings such as power, speed, and frequency in Lightburn to create markings that are both fast and legible
- Created more than 100 material samples, evaluating their aesthetics and durability with the use of barcode scanners, microscopes, and abrasion testing machines
- Evaluated 20+ lasers using a custom weighted matrix. Presented top candidates to a panel of ~10 senior engineers, along with a cost analysis tool for future engineers to analyze benefits of implementing a laser in assembly lines
- Used cost tool to recommend a handful of products that would benefit from this technology, some reducing task time by as much as 85%, and created a recommendation of how to integrate it into an existing assembly line

PROJECTS

Coast By Wire **Jan 2026 - May 2026**

- Researched a new drivetrain architecture where one-way bearings are replaced with an electromagnetic clutch and custom software, allowing CEV's Urban Concept vehicle to both coast and drive in reverse
- Designed a benchtop rig in Autodesk Inventor and a Simulink model to test the system and its controls
- Predicted changes in efficiency performance based added mass and increased power draw from the EM clutch

Cube Collecting Robot, Team Circuit Breakers **Jan 2025 - May 2025**

- Collaborated with two other students to design an autonomous robot to collect cubes in a 60-second match

- Utilized Arduino programming to control 2 H-bridge circuits for our differential drive system
- Designed a gravity deployed arm in Fusion 360 that falls at the start of the match using a calibrated “twitch”

HydroClip

Jan 2025 - May 2025

- Created a backpack mounted water bottle holder for easy access while walking
- Designed a “snap in place” buckle mechanism in Fusion 360 to secure the water bottle in place and prevent spills
- Split the buckle into parts, 3D printed, and glued pieces together to maximize strength in the direction of bending

DreamRide

Jan 2023 - May 2023

- Worked with two peers to design and build an Arduino-powered infant sleep system using a repurposed car seat, integrating four vibration motors and a speaker module to simulate the motion and sounds of a moving vehicle.
- Implemented user controls with potentiometers for adjustable vibration intensity and sound selection.

RESEARCH EXPERIENCE

Hasani Lab, Graduate Researcher

Sep 2026 - Present

- Developing a procedure to create microscopic channels on various media with an ultraviolet laser for use in immune cell activation via mechanical stimulation
- Designing a platform to condition immune cells continuously

CAMPUS INVOLVEMENT

Humor Us!, Cornell University, *Member / Social Media Coordinator/Vice President*

Sep 2023 - Present

- Write and perform a live sketch comedy show once a semester
- Manage the group’s Instagram page and YouTube and create flyers for shows
- Manage logistics for the groups including room booking, prop management, recruitment planning

Donlon Hall Council, Cornell University, *President*

Sep 2023 - May 2024

- Planned social events for Mary Donlon Hall, a freshmen residence hall with 500+ residents

Grinspoon Hillel, Organizational Design Intern

Aug 2024 - Dec 2024

- Organized one-on-one meetings with Jewish students to determine how Hillel can better serve the community

SKILLS

- **CAD:** Autodesk Inventor | Autodesk Fusion | PTC Creo
- **Programming:** MATLAB | Java | Python | Arduino IDE | ROS
- **Data Management/Automation:** Excel | Power BI | Power Automate | Power Apps | SharePoint List
- **Simulation:** Simulink, ANSYS Mechanical | Altair HyperMesh | Altair Optistruct | GENESIS
- **Manufacturing:** 3D Printing | Manual Mill | Manual Lathe | TRAK lathe | CAM | Laser Cutting/Engraving
- **Lab Equipment:** LabVIEW | Microscope
- **Creative:** iMovie | Canva | Acting | Writing
- **Languages:** Hebrew (fluent)

INTERESTS

- Sustainable energy | electric vehicles | biomedical devices and instruments | sketch comedy | video editing | robotics | one-pot meals